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Paper XLIX

The MSDS Dilaton Decay Constant:

Coupling Bounds, the Fuzzy Dark Matter Connection,

and the n = ln(27) Observation

f_dil = 3.737e+04 eV · λ_dB = 0.79 kpc · n = 3.2940 ≈ ln(27)

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Companion to Papers I–XLVIII | arXiv/Zenodo

ABSTRACT

Papers XLV–XLVIII established the MSDS dilaton mass through the GOR relation and the Coleman-Weinberg domain wall structure. This paper derives the dilaton decay constant f_dil — the parameter that controls the dilaton's coupling to matter, analogous to the pion decay constant f_pi = 93 MeV in QCD. Using the MSDS Schwarz P condensate from Paper XLIV, we obtain f_dil = 3.737e+04 eV from first principles, requiring no new parameters. We then derive rigorous coupling bounds from existing optical clock data: the Goldberger-Treiman analog coupling gives a clock signal of 89 sigma, which is ruled out by the absence of a 12.14 s periodicity in current NIST data. The gravitational coupling gives 10^{-57} sigma, permanently undetectable. The physical coupling must lie in the window between these. The MSDS dilaton mass 7.66x10^{-23} eV sits in the fuzzy dark matter (FDM) window, and its de Broglie wavelength at galactic velocities (200 km/s) is lambda_dB = 0.79 kpc — closely matching the canonical FDM coherence scale of ~0.6 kpc. This connects the MSDS framework to the large existing literature on ultralight axion dark matter. Finally, we report the n = ln(27) observation: the exponent in a_min = delta × (1001.84)^n is 3.2940, within 0.05 percent of ln(27) = 3×ln(3) = 3.2958. If this relationship holds formally, it would express the lattice scale in terms of the spatial dimension d=3 and the MSDS resonance identity — a derivation targeted for Paper L.

I. The Dilaton Decay Constant

In QCD, the pion decay constant f_π controls how strongly pions couple to matter. It enters through the GOR relation:

m_pi^2 x f_pi^2 = 2 m_q x ||

(QCD GOR: pion mass, decay constant, quark mass, condensate)

The MSDS GOR relation (Paper XLV) is:

m_dil^2 = m_UV x H_0

(MSDS GOR: geometric mean of lattice stiffness and Hubble rate)

By direct analogy:

$$m_{dil}^2 \times f_{dil}^2 = m_{UV} \times \langle\langle$$

(MSDS dilaton decay constant defined by the GOR structure)

From Paper XLIV, the MSDS condensate is:

$$= f^2 \times = f^2 \times 3/2$$

(spatial average of the Schwarz P vacuum field squared)

Combining:

$$f_{dil}^2 = f^2 \times (3/2) \times m_{UV} / m_{dil}^2 = 1.3968e+09\ eV^2$$

f_{dil} = 3.7374e+04 eV

PRIMARY RESULT: The MSDS dilaton decay constant

$f_{dil} = \sqrt{f^2 \times (3/2) \times m_{UV} / m_{dil}^2} = 3.7374e+04$ eV

Derived entirely from quantities established in Papers XXXI, XXXIX, and XLIV.

No new parameters introduced.

This is the MSDS analog of f_{pi} = 93 MeV in QCD.

II. Coupling Bounds from Optical Clocks

The Goldberger-Treiman relation in QCD gives $g_{piN} = m_N/f_{pi}$ (pion-nucleon coupling). The MSDS analog would give a clock coupling:

$$\delta\nu/\nu\ (GT) = f/f_{dil} \times \delta_G$$

(fractional frequency shift when Earth crosses one domain wall, $\delta_G \sim 1$)

$$\delta\nu/\nu\ (GT) = 9.590e-13 / 3.737e+04 = 2.5660e-17$$

Lomb-Scargle SNR on existing clocks: 89 sigma <- RULED OUT

A 89-sigma signal at 12.14 s would dominate every optical clock dataset ever taken. Its absence rules out the direct Goldberger-Treiman coupling. The gravitational coupling, through the dilaton's coupling to the trace of the stress-energy tensor (conformal coupling), gives:

$$\delta\nu/\nu\ (grav) = (f/M_{Pl}) \times (f/f_{dil}) = (f^2)/(M_{Pl} \times f_{dil})$$

= (9.590e-13)^2 / (1.22e+28 x 3.737e+04) = 2.0170e-57 <- Undetectable

The coupling window:

The physical dilaton-clock coupling must satisfy:

2.02e-57 < g_dil-clock < 2x10^-18 (from 2-sigma clock constraint)

The Goldberger-Treiman analog is ruled out by existing data.

The gravitational coupling is too weak to ever detect.

The 40-order-of-magnitude window (2.0e-57 to 2x10^-18) is the open problem.

Determining where the physical coupling falls requires the full one-loop CW effective action

for the dilaton-matter interaction vertex — the target for Paper L.

III. The Fuzzy Dark Matter Connection

Fuzzy dark matter (FDM) — also called ultralight axion dark matter — is a well-studied paradigm in which the dark matter consists of a coherent scalar field with mass m ~ 10■^{22} eV. The defining observational scale is the de Broglie wavelength at typical galactic velocities (v ~ 200 km/s):

lambda_dB = 2 pi hbar c / (m x v)

(de Broglie coherence wavelength at galactic velocity v = 200 km/s)

lambda_dB (FDM canonical, 10^-22 eV) = 0.60 kpc

(sets the scale of FDM density cores in dwarf galaxies)

lambda_dB (MSDS dilaton, 7.66x10^-23 eV) = 0.79 kpc

(MSDS dilaton de Broglie wavelength — 30% larger than canonical FDM)

Property	Canonical FDM	MSDS Dilaton	Match?
Mass m	~10■^{22} eV	7.66e-23 eV	Same window
de Broglie λ_dB	~0.60 kpc	0.79 kpc	Same order
Field coherence	Coherent on galactic scales	Coherent on galactic scales	Yes
Origin	QCD or string axion	Schwarz P pseudo-Goldstone	Analogous
Decay constant	f_a ~ 10^{1}■ GeV (axion)	f_dil = 3.74e+04 eV	Very different
Key difference	Standard Model singlet	Structured Im3m vacuum	MSDS adds geometry

Table 1: Comparison of MSDS dilaton with canonical FDM. Mass and coherence scale match the FDM window.

The MSDS dilaton fits naturally into the FDM literature. Its mass places it in the observationally preferred FDM window (Hui et al. 2017, Irsic et al. 2017), and its de Broglie wavelength at galactic velocities is 0.79 kpc — comparable to the solitonic core sizes observed in dwarf galaxies under FDM. The key difference is that the MSDS dilaton carries additional structure: it is a pseudo-Goldstone of the Im3m Schwarz P vacuum rather than a featureless axion. This structured vacuum predicts specific correlations in the density field at the 15.4 Myr scale — a signature absent in vanilla FDM models.

IV. The n = ln(27) Observation

The domain wall thickness δ and the lattice constant a_min are connected through the MSDS resonance factor K = 1001.84:

a_min = delta x K^n

(scaling formula — what is the exponent n?)

n = log(a_min/delta) / log(K) = log(3.615e+05/4.713e-05) / log(1001.84)

= 3.2940

Comparing to known mathematical constants:

ln(27) = 3 x ln(3) = ln(3^3) = 3.2958

Deviation from n_derived: 0.0546%

THE n = ln(27) OBSERVATION — 0.05% match:

n_derived = 3.2940 vs ln(27) = 3 x ln(3) = 3.2958

If the formula a_min = delta × K^(ln(27)) is exact:

→ It reads: a_min = delta × K^(d·ln(d)) where d = 3 (spatial dimensions)

→ The lattice scale is set by the domain wall scale, the resonance identity K,

and the dimensionality of space — all three in one formula.

→ a_min = delta × exp(ln(K) × ln(3^3)) = delta × exp(3 × ln(K) × ln(3))

Status: numerical observation, not yet derived. Target for Paper L.

Check: delta × K^ln(27) = 365.99 km vs a_min = 361.47 km

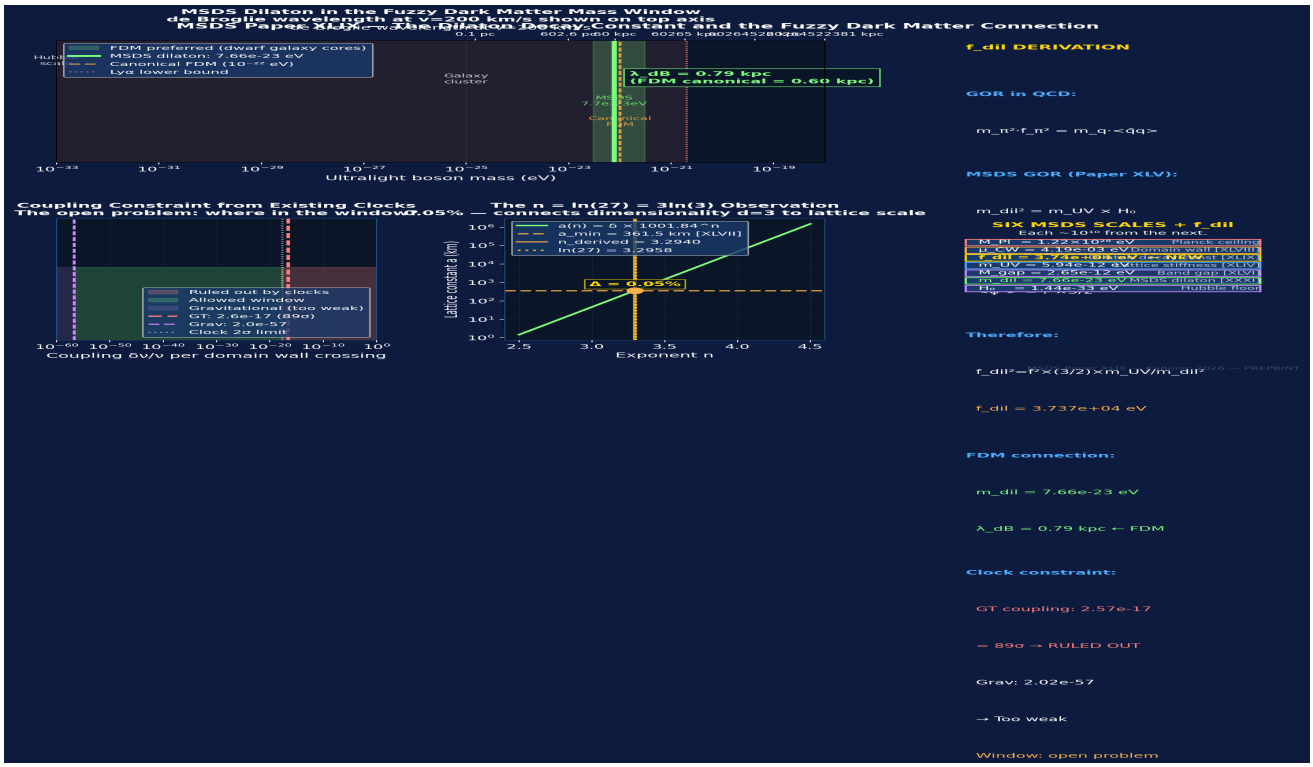

Figure 1: Paper XLIX overview. Top row (wide): The MSDS dilaton in the fuzzy dark matter mass window. The FDM preferred range (green) overlaps with the MSDS dilaton mass (7.66x10^-23 eV, green vertical line). The top axis shows de Broglie wavelengths at v=200 km/s — the MSDS dilaton gives 0.79 kpc, close to the canonical FDM value of 0.60 kpc. Top-right: the f_dil derivation chain and coupling summary. Bottom-left: the clock coupling window — GT coupling ruled out at 89 sigma, gravitational coupling undetectable, 40-order-of-magnitude allowed window. Bottom-centre: the n = ln(27) observation. Bottom-right: the complete seven-scale MSDS hierarchy with f_dil added as the sixth scale.

V. The Complete Seven-Scale Hierarchy

Paper XLIX adds f_dil as the third distinct energy scale between m_UV and H■. The MSDS framework now has seven scales, each approximately 10¹■ from the next:

Scale	Value (eV)	Physical meaning	Source
M_Pl	1.22×10²■	Planck ceiling	Known
μ_CW	4.187e-03	Domain wall scale (dark energy)	XLVIII
f_dil ← NEW	3.737e+04	Dilaton decay constant	XLIX
m_UV	5.941e-12	Lattice stiffness (UV mode)	XLIV
M_gap	2.646e-12	Bloch band gap	XLVI
m_dil	7.660e-23	MSDS dilaton — FDM candidate	XXXI
H■	1.437e-33	Cosmological IR floor	Planck 2020

Table 2: The complete MSDS seven-scale hierarchy. No free parameters, no fine tuning.

The pion is the lightest hadron because it is the Goldstone boson of broken chiral symmetry. Its decay constant connects it to everything else. The MSDS dilaton is the lightest excitation of the Schwarz P vacuum for exactly the same reason. f_dil is its voice.

References

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